

Review retrospective:
harmonic Laplace localisation, two dead opens, and an orphaned file

Claude Opus 4.8 (1M ctx), reviewer GPT-5.5 Pro

2026-06-05

Abstract

A soundness review of `Laplace/OneD/Localisation.lean`: the harmonic-tail localisation lemmas — for $L_\lambda(x) = (\lambda/2)x^2$, the contribution to $\int x^n e^{-tL_\lambda}$ from outside $(-\delta, \delta)$ is bounded by $C(\delta, \lambda, t) e^{-(\lambda t)\delta^2/2}$, which is $o(t^{-N})$ for every N . Result: **a clean fidelity pass, two dead opens removed**, plus a report-only observation that the file is now an orphan.

1 Setting

This file packages the `TailBound` Gaussian-tail estimates into harmonic-Gibbs localisation lemmas — the ingredients for controlling the away-from-origin part of the Stage-2 anharmonic covariance integrals. It is the last file of the deepest 1D foundation layer, reviewed after its dependency `Harmonic` and its (former) consumer `Anharmonic`.

2 What was reviewed

Five public theorems (zeroth/first/second-moment half-line tails and the zeroth/second two-sided tails) and a private even-symmetry helper. The half-line tails rewrite the integrand to rescaled-Gaussian form and delegate to `TailBound` ($b = \lambda t$); the two-sided tails double the half-line bound via the even-symmetry helper.

3 Fidelity / soundness findings

Sound; no mismatch. The integrand rewrite $e^{-t(\lambda/2)x^2} = e^{-(\lambda t)x^2/2}$ is correct; the second-moment by-parts form $\delta e^{-(\lambda t)\delta^2/2}/(\lambda t) + (1/\lambda t) \int_{x>\delta} e^{-(\lambda t)x^2/2}$ is bounded by the zeroth tail to give the stated two-term bound; the even-symmetry helper correctly maps $\text{lho}(-M) \rightarrow \text{lo} M$ for the even integrand $x^{2n} e^{-tL_\lambda}$, and is used only for the zeroth ($n = 0$) and second ($n = 1$) tails — consistent with the odd first-moment tail not getting a two-sided version. The two-sided factors of 2 are right.

One report-only, project-level observation (not a fidelity defect, not acted on): **this file is now an orphan**. Nothing imports it — the `Anharmonic` review removed the (genuinely unused) `import Laplace.OneD.Localisation`, and no other file imports it. The `harmonic_tail_*` lemmas are therefore landed-but-unconsumed; they were written “for the integrals we need in Stage 2”, so either the Stage-2 covariance proof will consume them or they become dead-file candidates. Flagged for the user; the lemmas themselves are correct and worth keeping until Stage 2 lands.

4 Simplifications applied

Two dead opens removed. `open Real` is dead (every `exp` in code is `Real.exp`; there is no $\pi/\text{sqrt}/\sqrt{}$ — these are tail bounds, no $\sqrt{2\pi}$; the only bare `exp` is in comments), and `open scoped Nat` is dead (no double-factorial; the `Nat.mul_zero/Nat.mul_one` uses are qualified). `open MeasureTheory` (bare $\int$) and `open Set` (bare `Ioi/Iio/Iic`) stay. Rebuild-arbitrated green, no warnings, six declarations byte-identical, GPT-affirmed.

Unlike the Harmonic review’s reverted block, removing *opens* carries no unused-binder cascade risk: an open is pure name resolution, so the only failure mode is an “unknown identifier” the rebuild catches immediately. GPT’s other suggestions (factor the three duplicated integrand-rewrite *haves* into a helper, golf the `_compl` ring rewrites, compress the second-moment inequality transport, rename locals) were declined as helper-extraction / tactic-golf.

5 Disagreements and open questions

None on the mathematics. The orphan status is surfaced for the user (follow-up, not a defect).

6 Lessons

- **Dead-open removal is the lowest-risk dead-code class.** An `open` binds no variables and contributes no proof term — it only affects name resolution. So removing a truly-dead open cannot trigger the unused-binder lint cascade that made the Harmonic file’s dead proof-block unsafe; the rebuild either errors (open was needed) or is clean (it was dead). Two dead opens here, both clean.
- **A review can surface a file becoming an orphan as a side effect of a sibling review.** Removing a foundation file’s unused import (`Anharmonic` $\rightarrow$ `Localisation`) can leave the imported file with no consumers. That is worth recording — not as a fidelity problem, but as a coverage/architecture signal that the file is either pending a future consumer or is itself dead weight.

7 Follow-ups

- **Orphan resolution (user decision):** either the Stage-2 anharmonic covariance proof should consume `Localisation`’s `harmonic_tail_*` lemmas, or — if Stage 2 takes a different route — the file is a dead-file candidate. Not a review-tide call.
- This completes the deepest 1D foundation layer (`Harmonic`, `Localisation`, `Anharmonic`, `Rescaling`, `GaussianMoments` all reviewed). Remaining tractable 1D files: the `J-function` and `Quartic/Sextic/Monomial` families; the `Multi/Covariance*` trio remains the large sectioned-approach target.